Why Are Batters Evaluated by Plate Appearance and Pitchers by Inning?

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Brief History of Pitching Stats

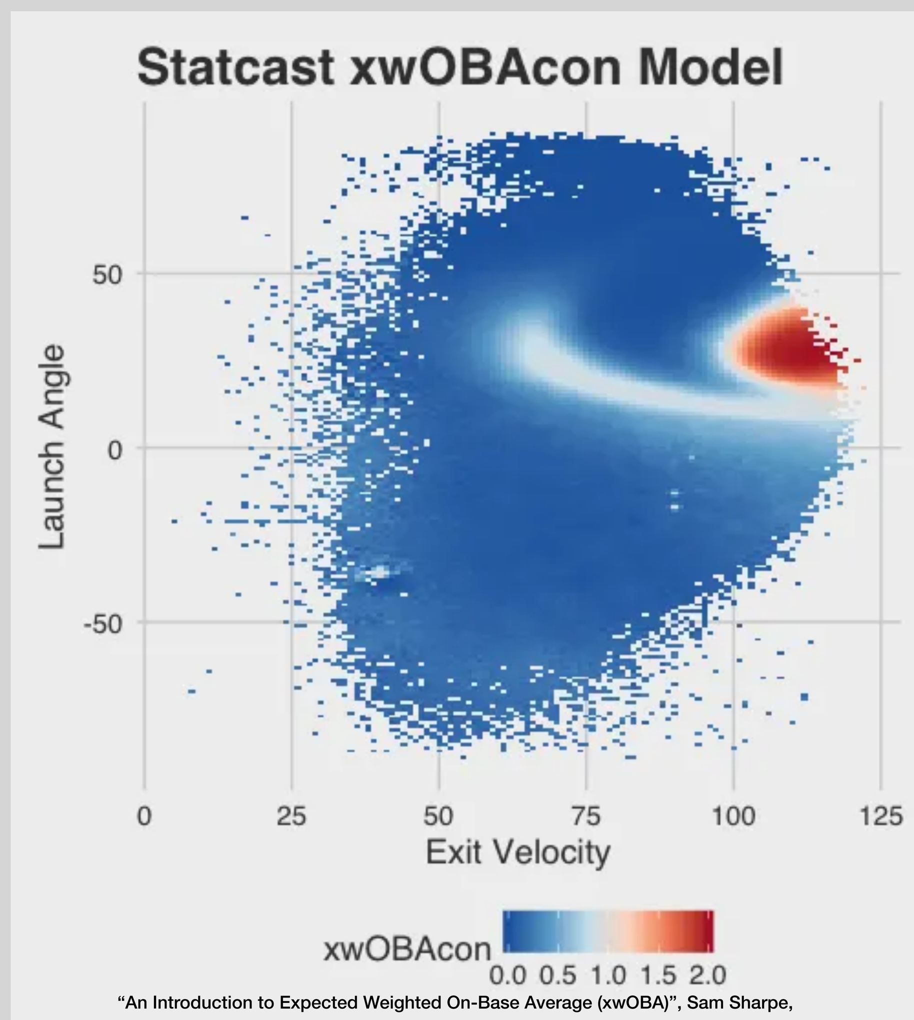
- 19th century: Henry Chadwick, ER per game
- 1912: John Heydler, ERA (earned run average), calculated per 9 innings, adopted by NL
- Long period of ERA, wins, and winning percentage
- 1999: Vöros McCracken, Defense Independent Pitching Statistics (DIPS)
- Early 2000s: Tom Tango, Fielding Independent Pitching (FIP)
- 2016: Statcast, wOBA (weighted on-base average) and xwOBA (expected wOBA per exit velocity and launch angle)
- Is that the end of the story?

Roadmap for this talk

- We look at common estimators for next year's ERA
- Current ERA is a weak predictor; other estimators like FIP and K%-BB% do a better job of predicting future ERA
- But how predictable are these estimators, and why are some better than others?
- It's hard to disentangle all the inputs into a pitcher's performance, but we can isolate sampling errors by freezing all the underlying stats (e.g., ERA, K%, wOBA)
- Each estimator's component variances add to the overall sample variance (standard deviation or noise)
- Sample variance is *why* ERA is a weak predictor and why lower variances make other estimators better
- The sample variances explain *about half* of the estimation error independent of pitcher quality, defense, etc.
- For example, xwOBA-2*K% has a stronger correlation than FIP, K%-BB%, or SIERA

Common Estimators for Pitching:

- ERA: Earned run average
- FIP: Fielding Independent pitching
  - Uses three true outcomes (TTO) :HR, K, BB+HBP)
- SIERA: Skill-interactive Earned Run Average
  - Nonlinear interaction of TTO, ground balls, and flyballs
- wOBA: Weighted On-Base Average
  - Proportional values for each hit type
- xwOBA: expected wOBA
  - wOBA for launch angle and velocity



Metric Comparison from Year to Year

	Correlation to Next Year ERA	Correlation to Current Year ERA
ERA	0.28	
K%-BB%	0.40*	0.54*
FIP	0.34	0.79
xFIP	0.37	0.60
SIERA	0.38	0.61
wOBA	0.32	0.90
xwOBA	0.35	0.77
xwOBAcon	0.11	0.63

* Only the absolute value of the correlation is relevant

Shouldn't a metric that is predictive for the following year also be good for the current year? One reason the don't is they include different things:

	ERA	K%-BB%	FIP	wOBA	xwOBA	SIERA
"True" pitching merit						
Three true outcomes		K, BB		BB, HR	BB,HR	
Batted ball outcome						
Quality of contact						
Ordering of events						
Earned & unearned						
Fielding						
Park effects						
Statistical variance						

To isolate statistical variance from sampling, we can freeze a pitcher's underlying metrics (e.g., ERA, K%, BB%, wOBA). For the sake of this talk, we used league average values:

Runs/Inning (2000-2025)

Runs	Innings	% of IP
0	157,231	74.69%
1	31,453	13.63%
2	14,894	6.45%
3	6,864	2.97%
4	3,000	1.30%
5	1,269	0.55%
6	559	0.24%
7	218	0.09%
8	86	0.04%
9	39	0.02%
10	18	0.01%
11	6	0.00%
12	1	0.00%
13	2	0.00%
14	1	0.00%

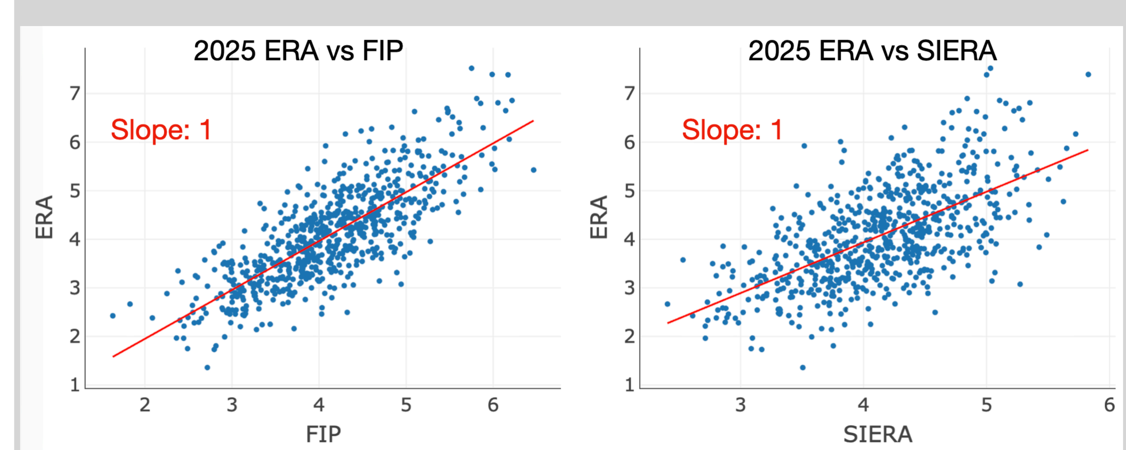
Outcome Frequency (2025)

	2025 Amount	2025 Pct/ PA
BB	15,379	8.4%
IBB	556	0.3%
HBP	1,928	1.1%
1B	26,115	14.3%
2B	7,745	4.2%
3B	628	0.3%
HR	5,650	3.1%
K	40,645	22.3%
PA	182,926	
IP	43,075	
lgERA	4.15	

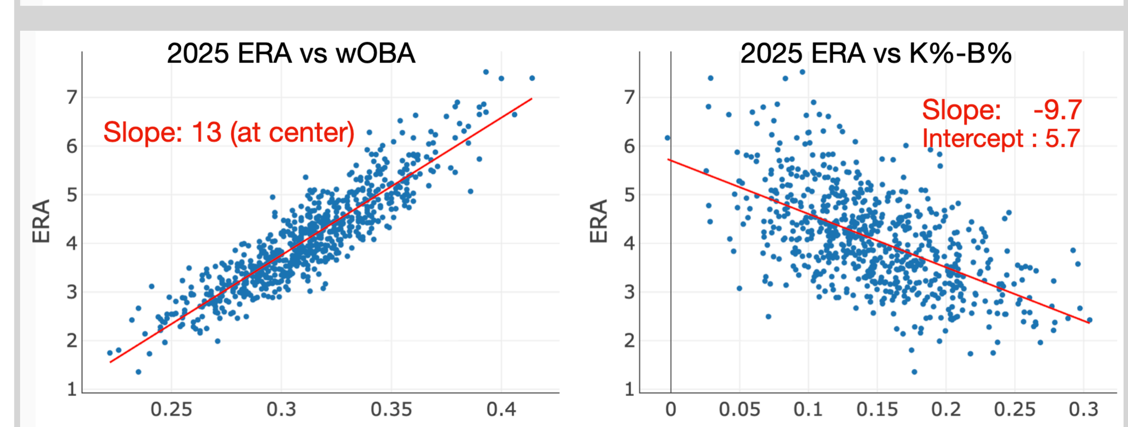
This means we're gong to look at the performance of the same pitcher assuming the same ballpark, the same fielders, and the same underlying average metrics.

We'll take 100,000 random draws for "average" 140 inning seasons, using innings (for ERA) or 630 plate appearances (for everything else, using an average of 4.5 plate appearance per inning, i.e. a WHIP of 1.5). The parameter likelihoods can exactly calculate the sample variance which validated the results (see the "Math Stuff" to the right).

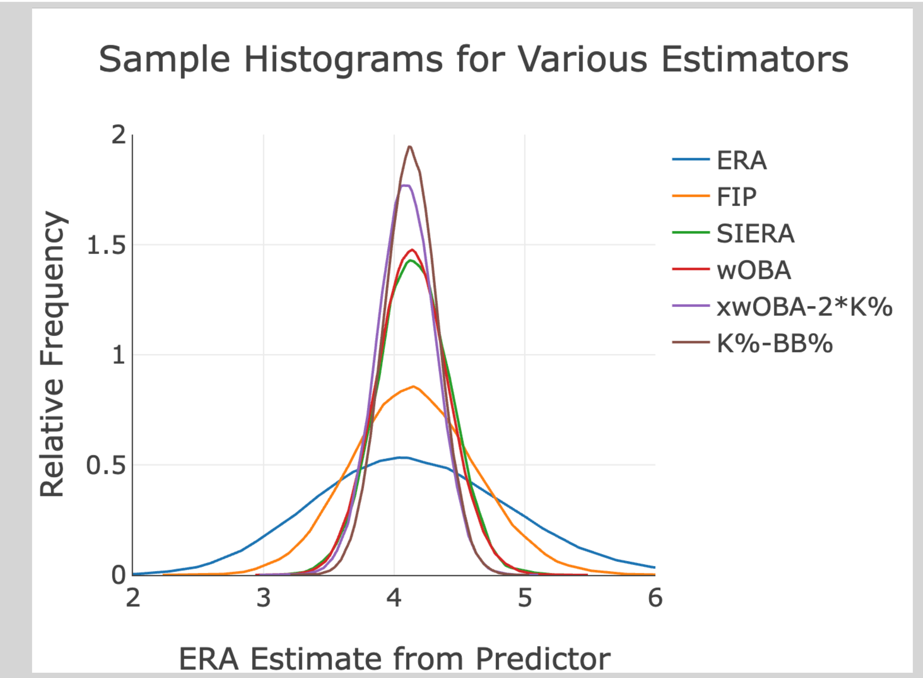
For common comparison, we use the ERA prediction from each estimator. This is determined by doing a linear fit of pitcher data for pitcher with at least 140 inning between 2021-2025.



Both scaled the same as ERA by design



Parameter variance for the ERA estimates scale by the square of the product of slope and the parameter weighting. The overall variance is the sum of the component variances minus any cross-correlation effects.



Right off the bat, ERA is a poor choice because sample variance is inversely related to the number of samples and 140 IP is less than 630 PAs. FIP is worse than K%-BB% because its weighting of HRs doubled its variance.

Variance and Future Correlation Coefficient of Estimators

Estimator	Sampling Std. Dev. (Runs)	Next-year Correlation Coefficient
ERA	0.75	0.28
K%-BB%	0.21	0.40
FIP (xFIP)	0.47	0.34
SIERA	0.27	0.37
wOBA (xwOBA)	0.27	0.38
xwOBA-2*K%	0.24	0.42

Correlating one noisy estimate (next year's ERA) with another noisy estimate will add the variances, hence low variance estimates perform better. As an example, xwOBA-2*K% performed better than the other estimators. Its variance is almost as good's K%-BB%, but it includes the effects of all plate appearances (not just BBs) without overweighting HRs like FIP and driving up the variance.

Key Takeaways:

- ERA is very noisy even with "known" underlying stats
  - Fewer samples and much higher variance
- Estimator variance is very important
  - FIP suffers from high variance due to over-weighting home runs
- e.g., xwOBA-2*K% has a stronger correlation with future ERA
  - Contains more of the underlying quality of contact than K%-BB%
  - Controls variance better than FIP
- Further research has to include sample variance along with biases and nonlinearities
- Given ERA's sampling variance, it might not be the best thing to match against anyway

Side Notes (Math stuff)

$$FIP = \frac{13*HR + 3*(BB+HBP) - 2*K}{IP} + FIP_c$$

$$FIP_c = ERA_{lg} - \frac{13*HR_{lg} + 3*(BB_{lg}+HBP_{lg}) - 2*K_{lg}}{IP_{lg}}$$

$$xFIP = \frac{13*FlyBalls*HR_{lg}/FlyBall + 3*(BB+HBP) - 2*K}{IP} + FIP_c$$

$$wOBA = \frac{0.690\cdot uBB + 0.722\cdot HBP + 0.888\cdot 1B + 1.271\cdot 2B + 1.616\cdot 3B + 2.101\cdot HR}{AB+BB-IBB+SF+HBP}$$

wOBA coefficients change every year

$$xwOBA = \frac{\left(\sum xwOBAcon\right) + 0.690(BB-IBB) + 0.722\cdot HBP}{AB+BB-IBB+SF+HBP}$$

Variance for Sum of Random Discrete Variables

- A discrete random number with likelihood p has variance $p(1-p)$
- For the sum of n samples, variance is $p(1-p)/n$
- For estimate $\frac{1}{n}\sum_i w_i p_i$, the variance is $\sigma^2 = \frac{1}{n}\sum_{i,j} C_{i,j}$ where $C_{i,i} = w_i^2 p_i(1-p_i)$ and $C_{i,j\neq i} = -w_i w_j p_i p_j$

Relation Between Correlation Coefficient and Noise

$$r = \frac{var\left(ERA - lgERA\right)}{var\left(ERA - lgERA\right) + var(total\ noise)}$$

$$var(total\ noise) = var(ERA - lgERA)\frac{1-r}{r}$$

$$var(total\ noise) = var(ERA - lgERA)\frac{1-r}{r} = 0.8\frac{1-0.4}{0.4} = 1.2$$

Total sampling variance matching a parameter to the next year's ERA is the sum of the sampling errors for the ERA and the estimator (0.567+0.09=0.657).

About half of the overall variance is due to ERA sampling noise!

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